

DAY	TOPIC	Project/Exercise
. 1	• Python Introduction and Environment	Install Anaconda, IDEs setup
. 2	Setup • Python Syntax, Variables, Data Types	• Simple math operations
. 3	Operators and Expressions	Logical operation mini exercises
. 4	Conditional Statements (if, else, elif)	Create a grading system
. 5	Loops (for, while, nested loops)	Pattern generation
. 6	Functions, Lambda, Map, Filter	Temperature converter using
. 7	Data Structures (List, Tuple, Set, Dictionary) • String Handling and Operations	functions Simple address book
. 8	File Handling and Exceptions	Text manipulation exercises
. 9	• OOP Concepts: Classes and Objects	File reader/writer tool
. 10	Introduction to Machine Learning	Bank management system
. 11	Supervised vs Unsupervised	• Theory session
. 12	Learning Data Preprocessing and EDA	Examples discussion
. 13	Linear Regression Concept and	• Titanic dataset analysis
. 14	Code Classification: Logistic Regression	• Predict house prices
. 15	• KNN and Decision Trees	Email spam classification
. 16	Random Forest and Model	Iris dataset classification
. 17	Evaluation Clustering: K-Means and	Precision, Recall, F1-Score
. 18	Hierarchical Dimensionality Reduction: PCA	practice • Customer segmentation
. 19	• Mini Project 1 Start: Customer Churn	• Visualize dataset post PCA
. 20	Prediction • Mini Project 1 Development	Churn dataset EDA and
. 21	• Mini Project 1 Enhancement	preprocessing Model building and testing
. 22	• Mini Project 1 Submission	• Hyperparameter tuning
. 23	Introduction to Deep Learning	Presentation and feedback
. 24	Neural Networks Components	• Understanding Deep Neural
. 25	Tensorflow and Keras Basics	Networks • Activation functions explanation
. 26	Build Simple ANN Model	Hands-on basic model creation
. 27	Forward Propagation and	• Digit recognizer project
. 28	Backpropagation Explained Building CNNs for Image	• Theory discussion
. 29	• Mini Project 2 Start: Disease	• MINIST dataset CNN
. 30	Prediction • Mini Project 2 Development	• Prepare healthcare dataset
. 31	• Mini Project 2 Enhancement	• Train and evaluate healthcare
. 32	• Mini Project 2 Submission	prediction model • Model tuning and analysis
. 33	Introduction to Generative AI	• Presentation and demo
. 34	• GANs Introduction (Generative	Generative AI use cases
. 35	Adversarial Networks) Understanding Generators and	Concept and architecture
. 36	Discriminators Building First Simple GAN	Illustrative examples
. 37	• VAEs (Variational Autoencoders)	Generate handwritten digits
. 38	• Diffusion Models	Basic concept
. 39	• Large Language Models (LLMs)	• Theory and application
. 40	• RAG Architecture Basics (Retrieval	Introduction to LLMs
. 41	Augmented Generation) • Building a RAG-Based QA System	Theory and examples
. 42	• Using Huggingface Transformers	Dataset preparation and RAG
. 43	Prompt Engineering Basics	integration Load pre-trained models
. 44	Fine-tuning Small LLMs	Few-shot, zero-shot prompting
. 45	Gen AI for Text Generation	• Theory and demo
. 46	Gen AI for Image Generation (DALL-	Text completion project
. 47	E, Stable Diffusion) • Application: Building a Chatbot using	Image generation exercises
. 48	LLM • Application: AI Content Writer	• Chatbot flow design
. 49	Sentiment Analysis using LLM	Mini project idea discussion
. 50	• Mini Project: RAG + LLM Application	Train and deploy sentiment
. 51	Development • Mini Project Development	analyzer Start working on RAG-based
. 52	• Mini Project Testing	chatbot Continue RAG/GenAI
. 53	• Mini Project Presentation	application Deploy and analyze
. 54	Revision and Interview Questions	performance • Showcase Gen AI solution
. 55	Resume Preparation and Career	Mock technical interview
. 56		Portfolio building